



# Hypertension

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CPhA acknowledges the contribution of Dr. Raj Padwal as a previous author of this chapter. CPhA strives to provide guidance for health-care professionals on the appropriate care of all individuals. However, we recognize that in this chapter, our content is based on clinical research studies of patients who may not reflect the identity of the individual patient presenting. Other resources exist that may be helpful, such as the World Professional Association for Transgender Health's (WPATH) Standard of Care. When the terms "men," and "women" are used in the text, they are used to maintain accuracy with what is reported in the clinical literature. We are aware these terms are not inclusive of all patients.

## Introduction

High blood pressure (hypertension) is the leading cause of death or disability in the world.<sup>[1]</sup> Approximately 1.4 billion individuals, or 31% of adults, have hypertension globally.<sup>[2]</sup> In Canada, approximately 23% of adults, or 7.5 million individuals, have hypertension and nearly 65% have achieved target BP levels.<sup>[3]</sup>

Regulation of BP is complex and is controlled by several physiological systems. Important influencers include the autonomic nervous system, renin-angiotensin axis, kinin-kallikrein system, cardiac function and output, vascular function, and renal salt and water balance. Additional factors that commonly lead to BP variation include body position, emotional state, external temperature and activity. Two major drivers of elevated BP are obesity (especially in the young) and vascular stiffness. Elevated BP leads to CV and renal complications. Important sequelae are stroke, myocardial infarction, atrial fibrillation, heart failure, chronic kidney disease and dementia.<sup>[4]</sup> Consider screening adults for hypertension annually, more often in those with borderline-high BP or diabetes.

## Goals of Therapy

- Reduce the risk of premature cardiac, cerebrovascular, renal and other vascular morbidity and mortality
- Achieve BP targets in treated patients
- Minimize the adverse effects of antihypertensive therapy

## Investigations

- History:
  - duration of hypertension, usual BP level, any sudden change in severity of hypertension, and prior hospitalization or emergency department visit for hypertensive urgency or emergency
  - use of home monitoring (proper cuff size, validated device [available from [Hypertension Canada](#)]), home BP levels, noting any discrepancy between home and office values
  - history of antihypertensive drug use, reason for changing/stopping therapy, effectiveness, side effects and intolerance
  - drugs that may cause hypertension (see [Table 1](#))
  - drugs that may interact with antihypertensive drugs (those that induce or inhibit metabolism)
  - adherence with lifestyle recommendations and drug therapy
  - family history of hypertension, CV risk factors and premature CV disease
  - personal history of cigarette and alcohol use, usual physical activity, usual diet and sodium intake, current weight and recent weight change, waist circumference, diabetes, dyslipidemia, and hypertension during pregnancy
  - cerebrovascular, cardiac and peripheral vascular symptoms to assess for target organ damage
  - symptoms of secondary hypertension, which include, for example, pheochromocytoma (hyperadrenergic symptoms), hyper- and hypothyroidism, Cushing syndrome, renal/urinary symptoms or past history of renal disease

**Table 1: Drugs and Other Exogenous Factors That Can Induce or Aggravate Hypertension<sup>[5]</sup>**

|                                                                  |                                                                          |
|------------------------------------------------------------------|--------------------------------------------------------------------------|
| Alcohol (excessive use)                                          | Nonsteroidal anti-inflammatory drugs (NSAIDs) including COX-2 inhibitors |
| Bruton tyrosine kinase inhibitors, e.g., ibrutinib, zanubrutinib | Oral contraceptives and sex hormones                                     |
| Calcineurin inhibitors, e.g., cyclosporine, tacrolimus           | Salt (sodium—high intake)                                                |
| Corticosteroids and anabolic steroids                            | Selective serotonin reuptake inhibitors (SSRIs)                          |
| Erythropoietin and analogues                                     | Serotonin-norepinephrine reuptake inhibitors (SNRIs)                     |
| Licorice root                                                    | Stimulants, including cocaine                                            |
| Midodrine                                                        | Vasoconstricting, sympathomimetic decongestants                          |
| Monoamine oxidase inhibitors (MAOIs)                             |                                                                          |

- Diagnosis:<sup>[5][6]</sup>
  - diagnosis is immediate in the case of hypertensive emergencies and urgencies; this includes patients with hypertension that is compromising vital organ function (encephalopathy, cardiac or rapidly decreasing renal function), hypertension and a major artery dissection, or those with DBP ≥130 mm Hg
  - diagnostic process in nonurgent cases is summarized in [Figure 1](#); mandatory elements for accurate diagnosis include strictly following recommended techniques, using the mean of multiple readings for clinical decision-making, and using

- out-of-office measurement to rule in or rule out the diagnosis
- measurements using a validated electronic device are preferred to auscultation. Automated office measurement is the preferred method of measuring BP in the office or pharmacy setting. An automated office device automatically performs serial measurements (usually 3–6 depending on the device) and calculates the mean. Ensuring that the patient is unattended during automated measurement can minimize white coat effect. However, all in-office measurement is susceptible to white coat effect, and the method that most closely predicts CV risk is out-of-office measurement (home BP or 24-hour ambulatory BP monitoring) for diagnosis and follow-up
  - hypertension may be diagnosed if the mean BP is consistently  $\geq 130/80$  mm Hg (see [Pharmacologic Choices](#))
  - diagnosis through out-of-office measurement can be done by performing a 24-hour ambulatory BP monitoring study or a home BP series; ambulatory (continuous) monitoring is preferred and is considered the gold standard of BP measurement. The home BP series comprises 2 readings taken each morning and evening for 7 days (28 total readings); discard the first-day readings and use the mean of the remaining 24 readings
  - out-of-office measurement of BP can identify white coat hypertension and masked hypertension; regular home BP measurement can improve BP control and improve medication adherence
  - Physical exam:
    - fundi for hypertensive retinopathy
    - bruits and peripheral pulses for vascular disease and renovascular hypertension
    - edema and lung fields for signs of heart failure
    - heart sounds (fourth heart sound)
    - sustained and displaced apex for left ventricular hypertrophy
    - abdominal mass for polycystic kidneys and aortic aneurysm
    - neurologic exam for cerebrovascular disease
    - femororadial delay and murmur for aortic coarctation
  - Initial laboratory testing:
    - serum potassium, sodium and creatinine
    - urinalysis
    - urinary albumin and/or albumin-creatinine ratio in patients with diabetes
    - fasting glucose and/or HbA<sub>1c</sub>
    - total cholesterol, HDL-C, LDL-C, triglycerides (lipids may be measured in the fasting or nonfasting state)
    - standard 12-lead ECG
    - consider renin and aldosterone levels in medication-naïve patients
    - select patients should have additional testing (see [Table 2](#))

**Table 2:** Hypertensive Patients Requiring Additional Laboratory Testing<sup>[5]</sup>

| If these characteristics are present:                                                                                                        | Check for:              |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <ul style="list-style-type: none"><li>High serum creatinine (high normal in older patients)</li></ul>                                        | Renal disease           |
| <ul style="list-style-type: none"><li>Urinary albumin or protein</li></ul>                                                                   | Diabetes, renal disease |
| <ul style="list-style-type: none"><li>Paroxysmal and/or severe sustained hypertension refractory to usual antihypertensive therapy</li></ul> | Pheochromocytoma        |

| If these characteristics are present:                                                                                                                                                                                                                                                                                                                                                                                                                                              | Check for:           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <ul style="list-style-type: none"><li>Hypertension and symptoms suggestive of catecholamine excess (2 or more of headache, palpitations, sweating, etc.)</li><li>Hypertension triggered by beta-blockers, monoamine oxidase inhibitors, micturition or changes in abdominal pressure</li><li>Incidentally discovered adrenal adenoma</li><li>MEN 2A or 2B; von Recklinghausen neurofibromatosis or von Hippel-Lindau disease</li></ul>                                             |                      |
| <ul style="list-style-type: none"><li>Spontaneous hypokalemia</li><li>Profound diuretic-induced hypokalemia (<math>K^+ &lt; 3</math> mmol/L)</li><li>Hypertension refractory to treatment with <math>\geq 3</math> drugs</li><li>Incidental adrenal adenoma</li></ul>                                                                                                                                                                                                              | Hyperaldosteronism   |
| <p>Two or more of:</p> <ul style="list-style-type: none"><li>Sudden onset or worsening of hypertension in patients <math>&gt;55</math> y or <math>&lt;30</math> y</li><li>Abdominal bruit</li><li>Uncontrolled hypertension despite use of <math>\geq 3</math> drugs</li><li>Decreased renal function associated with use of an ACE inhibitor or ARB</li><li>Overt atherosclerotic vascular disease</li><li>Recurrent episodes of hypertension and flash pulmonary edema</li></ul> | Renovascular disease |
| <b>Abbreviations:</b> ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; MEN = multiple endocrine neoplasia                                                                                                                                                                                                                                                                                                                                                  |                      |

## Therapeutic Choices

### Nonpharmacologic Choices

- All individuals should be advised about a healthy lifestyle to prevent or control hypertension and CV disease (see [Table 3](#)).
- Weight loss of 4 kg or more if overweight (target BMI: 18.5–24.9 kg/m<sup>2</sup>; waist circumference  $<102$  cm in men and  $<88$  cm in women).
  - Healthy diet—high in fresh fruits, vegetables, soluble fibre and low-fat dairy products, low in saturated fats and sodium, e.g., [DASH eating plan](#).
  - Sodium intake target of  $<2000$  mg (88 mmol) per day.

- Dietary potassium intake increase (e.g., fruit and vegetable component of DASH eating plan) if the patient is not at risk of hyperkalemia. Risk factors include renin-angiotensin inhibitors or other agents that can increase potassium, chronic kidney disease and serum potassium >4.5 mmol/L.
- Regular, moderate-intensity cardiorespiratory physical activity for 30–60 minutes on most days.
- Low-risk alcohol consumption (0–2 standard drinks/wk).
- Smoke-free environment.

Table 3: Effect of Lifestyle Changes on Blood Pressure in Adults with Hypertension

| Intervention                | Recommendation                    | Change in Blood Pressure (systolic/diastolic) mm Hg |
|-----------------------------|-----------------------------------|-----------------------------------------------------|
| Reduction in sodium intake  | Reduce by 1800 mg (78 mmol)/day   | −5.8/−2.5                                           |
| Weight loss                 | Reduce by 4.5 kg                  | −7.2/−5.9                                           |
| Reduction in alcohol intake | Reduce by 2.7 standard drinks/day | −4.6/−2.3                                           |
| Exercise                    | 30–45 min, 3 times/wk             | −10.3/−7.5                                          |
| Dietary modification        | DASH eating plan <sup>[a]</sup>   | −11.4/−5.5                                          |

<sup>[a]</sup> Available from: [www.nhlbi.nih.gov/education/dash-eating-plan](http://www.nhlbi.nih.gov/education/dash-eating-plan). Adapted with permission from Campbell N. Canadian Hypertension Education Program. Brief overview of 2004 recommendations. *Can Fam Physician* 2004;50:1411-5.

Pharmacologic Choices

The choice of therapy is determined by the measured BP values:

- In patients with BP ≥140/90 mm Hg or SBP of 130–139 mm Hg who are at high CV risk (see Table 4), pharmacologic treatment is recommended in addition to nonpharmacologic measures.
- If the SBP is 130–139 and CV risk is not high, initiate health behaviour modification. Monitor BP and other risk factors, regardless of whether such a patient chooses to begin drug therapy, as generally the risks accumulate and BP increases over time. Reassess the need for pharmacotherapy every 6–12 months.

Table 4: Factors Indicating High Cardiovascular Risk <sup>[6]</sup>

|                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------|
| Established cardiovascular disease (coronary artery disease, heart failure, cerebrovascular disease, peripheral artery disease) |
| Diabetes mellitus (type 1 or type 2)                                                                                            |
| Chronic kidney disease (eGFR <60 mL/min/1.73 m <sup>2</sup> or albuminuria ≥3 mg/mmol)                                          |

10-year Framingham Risk Score ≥20%

Age ≥75 years

Consider performing a pregnancy test in patients of childbearing age prior to initiating antihypertensive drugs. Also consider discussing plans for future pregnancy to optimize medication selection.

Consider therapy for dyslipidemia if the patient meets the current Canadian criteria (see [Dyslipidemias](#)).

Drug Selection

In general, the reduction in CV risk depends more on the extent of the reduction in BP than on the specific antihypertensive medication.<sup>[9]</sup> The 2025 Hypertension Canada Primary Care guidelines recommend initial therapy as a half dose of a single pill combination of angiotensin receptor blocker and diuretic (see Figure 2), such as irbesartan/hydrochlorothiazide (starting dose 150 mg/12.5 mg). This can be doubled after 4–6 weeks if the patient has not reached their target BP of <130 mmHg systolic.<sup>[6]</sup>

About 70% of patients will require more than 1 antihypertensive agent to achieve BP targets.<sup>[10]</sup> If the goal BP is not achieved with moderate doses of a suitable first-line drug, add, rather than substitute, a second drug. Combining 2 drugs from different classes yields a 5 times greater incremental reduction in BP than doubling the dose of 1 drug and has fewer side effects.<sup>[11]</sup>

Dose titration to achieve goal BP should be done approximately every 4 weeks for all but those with severe hypertension, target organ damage or high CV risk for whom closer follow-up and more frequent dosage titration is required. Lack of control over BP is in most cases due to a failure to titrate therapy (adding drugs and/or increasing doses) in response to high office readings. Greater confidence in office readings can result from supplementing with home BP measurements or ambulatory 24-hour monitors. Generally, high readings should trigger a dosage increase, addition of another medication, investigations to identify the cause of the high readings or a follow-up appointment within 2–4 weeks to reassess BP.

Diuretics

Thiazide, or thiazide-like diuretics (e.g., **chlorthalidone**, **hydrochlorothiazide** or **indapamide**), are considered first-line therapy in combination with an angiotensin receptor blocker.<sup>[6]</sup>

Diuretics can cause hypokalemia that may be associated with adverse CV outcomes. Consider alternative first-line agents in those with or strongly predisposed to a serious arrhythmia, e.g., prolonged QT syndrome. Consider using a combination product to minimize the risk of hypokalemia (hydrochlorothiazide plus a potassium-sparing diuretic—**spironolactone**, **amiloride** or **triamterene**). Reserve the use of high doses (e.g., >25 mg/day of hydrochlorothiazide) for patients with resistant hypertension unresponsive to treatment with multiple drugs or secondary to renal impairment. Consider using a loop diuretic in patients with renal impairment. Diuretics may worsen dysglycemia, although CV outcomes in patients with diabetes who are treated with diuretics are similar to those treated with ACE inhibitors.<sup>[10]</sup>

Beta<sub>1</sub>-Adrenergic Antagonists

Beta<sub>1</sub>-adrenergic antagonists (**beta-blockers**) are not considered first-line therapy, except in patients who have another indication such as stable angina, heart failure or a history of MI. Beta-blockers are also useful in patients who have migraine headaches, tachycardia or essential tremor.

## Drugs That Act via the Renin-Angiotensin-Aldosterone System

The renin-angiotensin-aldosterone system (RAAS) plays a crucial role in modulating BP, kidney function, electrolyte balance, and vascular and cardiac structure. Drugs that act directly on this system include **ACE inhibitors, angiotensin receptor blockers, direct renin inhibitors** and **spironolactone**. Antihypertensive drugs that stimulate the RAAS axis (e.g., diuretics) are as effective as those that block this system in preventing CV events in patients with hypertension. However, some inhibitors of the RAAS do provide additional benefits in certain patients, including those with heart failure, diabetes and/or chronic kidney disease. ACE inhibitors, ARBs and direct renin inhibitors are contraindicated in pregnant patients.<sup>[12][13][14]</sup> Drugs from these classes should not be prescribed in those of childbearing potential unless the risks are carefully weighed and adequate measures are taken to prevent pregnancy (see [Choices during Pregnancy and Breastfeeding](#)).

### Angiotensin-Converting Enzyme Inhibitors

ACE inhibitors can be first-line agents (in a single pill combination with a thiazide diuretic) for patients with uncomplicated hypertension and for patients with diabetes, ischemic heart disease, recent MI, heart failure or chronic kidney disease. However, angiotensin receptor blockers are often better tolerated by patients, as cough is less frequent.

### Angiotensin II Receptor Blockers

Long-acting ARBs are first-line agents (in a single pill combination with a thiazide diuretic) for patients with uncomplicated hypertension, diabetes or ischemic heart disease.

### Direct Renin Inhibitors

**Aliskiren**, a direct renin inhibitor, prevents renin from converting angiotensinogen to angiotensin I. The drug has a long duration of action and may lower BP to the same extent as drugs from other antihypertensive classes. Large-scale trials examining morbidity and mortality endpoints are lacking. Aliskiren was withdrawn from the Canadian market in 2024 for commercial reasons.

## Long-Acting Calcium Channel Blockers

Long-acting dihydropyridine CCBs, such as amlodipine, should be added in addition to an angiotensin receptor blocker and diuretic combination as a third step in patients who have not achieved their target BP. *Short-acting formulations* of these agents (nifedipine) have caused an increase in CV events in randomized controlled trials and should *not* be used, except in the acute management of severe hypertension in pregnancy (see Hypertension and Pregnancy, [Management](#)). Older patients with isolated systolic hypertension are particularly responsive to CCBs.<sup>[10][15]</sup>

## Other Antihypertensive Drugs

In general, other classes of antihypertensive drugs should not be prescribed unless there are specific indications (see [Table 5](#)), contraindications or intolerance to first-line therapy, or a requirement for additional BP lowering in combination with first-line antihypertensive drugs. Medications that can be considered are shown in [Table 6](#).

## Adherence

Medication adherence should be assessed at each visit. Poor adherence to therapy is a major cause of poor BP control. Evening dosing has been suggested to provide better outcomes,<sup>[16]</sup> but these findings have been refuted by the TIME trial.<sup>[17]</sup> At this point, there appears to be no advantage in using bedtime dosing of antihypertensive medications routinely.

Patients may admit to poor adherence when questioned in a nonconfrontational manner, or it may be indicated by:

- Failure to keep scheduled appointments
- Poor BP control
- Lack of secondary physiologic effects, e.g., decreased heart rate on beta-blocker
- Failure to renew prescriptions on time
- Lack of awareness of usual pill-taking routine and prescriptions

Poor adherence can be prevented. Routine care should include the following:

- Ensure patients are well informed about hypertension and its treatment, preferably verbally and with patient information pamphlets (available on the [Hypertension Canada](#) website)
- Include family or social support in lifestyle modification
- Use a simplified regimen of long-acting, once-daily drugs, and prescribe formulations that contain 2 drugs in combination when appropriate
- Ensure patients can afford the prescribed drugs
- Advise patients to establish a daily routine for pill-taking, e.g., putting their pills by their toothbrush and taking them every morning prior to brushing

Treat poor adherence:

- Determine the reason for poor adherence and tailor advice or interventions to the cause
- Advise use of adherence-enhancing medication dispensers, e.g., dosette box
- Advise self-measurement of BP
- Consider assessing adherence with an electronic pill dispenser
- Advise home-monitoring of adherence with pill counts and marking on a calendar when the prescription needs renewing
- Consider regular telephone contact with patients, if feasible
- Enlist social work assistance

## Resistant Hypertension

Resistant hypertension is defined as BP that is above target despite treatment with 3 drugs, optimally dosed, 1 of which is a diuretic (typical “optimal” regimen is a long-acting ACE inhibitor or ARB, thiazide-like diuretic, and dihydropyridine CCB). Resistant hypertension has been estimated to affect about 10% of hypertensive patients, but the true prevalence is likely much lower because many patients have either white coat effect or are not adherent.<sup>[18]</sup> Hence, all patients with suspected resistant hypertension should undergo 24-hour ambulatory BP measurement (ABPM) and have adherence assessed. If there is no nocturnal dipping found on 24-hour ABPM, screening for sleep apnea is indicated and moving 1 or more medication doses to the evening may be beneficial. Other potential causes that should be looked for are secondary hypertension and renal dysfunction; in those with a poor response to an adequate combination of medications, consider the possibility of an “interfering lifestyle,” e.g., high dietary sodium intake. Therapeutic options as add-on therapy to the optimal 3-class regimen described above include spironolactone, amiloride, bisoprolol, doxazosin and clonidine.<sup>[19]</sup> Refer (to a hypertension specialist, nephrologist or internist) those who do not achieve BP targets with medication regimens you feel comfortable prescribing.

## Hypertensive Emergencies

It is uncommon for elevated BP alone, without new or progressive target organ damage, to require emergency therapy. Refer true hypertensive emergencies to experienced centres with facilities to continuously monitor BP. In stabilizing patients for transfer, the use of intermediate-acting drugs (e.g., **felodipine**) with close BP monitoring is generally safer than using short-acting drugs that can rapidly produce hypotension with complications.

**Table 5: Individualization of Antihypertensive Therapy**<sup>[5]</sup>

| Category                                                 | Risk Factor/Disease                                                                  | Initial Therapy                                                                                                                                                    | Second-Line Therapy                                                                                                                                      | Notes/Cautions                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hypertension without other compelling indications</b> | Diastolic ± systolic hypertension                                                    | ARB (or ACE inhibitor) plus thiazide diuretic single pill combination                                                                                              | Add amlodipine to ARB/diuretic combination                                                                                                               | Beta-blockers are not recommended as initial therapy in patients over 60 y.                                                                                                                                                                                                                                                                      |
|                                                          |                                                                                      | Consider statins in select patients<br><br>Consider initiating therapy with a combination of first-line drugs if SBP is ≥20 mm Hg or DBP is ≥10 mm Hg above target |                                                                                                                                                          | Avoid hypokalemia in those who are prescribed diuretics as monotherapy by using K <sup>+</sup> -sparing agents.<br><br>ACE inhibitors, ARBs and direct renin inhibitors are teratogenic. Marked caution is required if prescribing to patients of childbearing potential.<br><br>Combination of an ACE inhibitor with an ARB is not recommended. |
|                                                          | Isolated systolic hypertension without other compelling indications                  | Thiazide diuretic, ARB or long-acting dihydropyridine CCB                                                                                                          | Combinations of first-line drugs                                                                                                                         | See diastolic ± systolic hypertension above.                                                                                                                                                                                                                                                                                                     |
| <b>Diabetes mellitus</b>                                 | Diabetes mellitus with albuminuria, renal disease, CVD or additional CV risk factors | ACE inhibitor or ARB                                                                                                                                               | Addition of a dihydropyridine CCB is preferred over thiazide diuretics                                                                                   | A loop diuretic could be considered in hypertensive CKD patients with extracellular fluid volume overload.                                                                                                                                                                                                                                       |
|                                                          | Diabetes mellitus not included in the above category                                 | ACE inhibitor, ARB, long-acting dihydropyridine CCB or thiazide diuretic                                                                                           | Combinations of first-line drugs<br><br>If combination with ACE inhibitor is being considered, a dihydropyridine CCB is preferable to thiazide diuretics | Albuminuria is defined as an albumin to creatinine ratio (ACR) >2 mg/mmol.<br><br>Combination of an ACE inhibitor with an ARB is specifically not recommended.                                                                                                                                                                                   |

| Category                                           | Risk Factor/Disease          | Initial Therapy                                                                                                                                 | Second-Line Therapy                                                                                                                                                                                                                                                                                                                                                                                            | Notes/Cautions                                                                                                     |
|----------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Cardiovascular and cerebrovascular diseases</b> | Coronary artery disease      | ACE inhibitor or ARB; beta-blocker or CCB for patients with stable angina                                                                       | Long-acting CCB<br><br>When combination therapy is being used for high-risk patients, an ACE inhibitor/dihydropyridine CCB is preferred                                                                                                                                                                                                                                                                        | Avoid short-acting nifedipine.<br><br>Combination of an ACE inhibitor with an ARB is specifically not recommended. |
|                                                    | Recent MI                    | Beta-blocker and ACE inhibitor<br><br>(ARB if ACE inhibitor not tolerated)                                                                      | Long-acting CCB                                                                                                                                                                                                                                                                                                                                                                                                | Non-dihydropyridine CCBs should not be used in the presence of concomitant heart failure.                          |
|                                                    | Heart failure                | ACE inhibitor and beta-blocker<br><br>(ARB if ACE inhibitor not tolerated)                                                                      | ARB added to ACE inhibitor                                                                                                                                                                                                                                                                                                                                                                                     | Titrate doses of ACE inhibitors and ARBs to those used in clinical trials.                                         |
|                                                    |                              | Aldosterone antagonist in patients with a recent CV hospitalization, acute MI, elevated BNP, elevated NT-proBNP or NYHA class II to IV symptoms | Angiotensin receptor-neprilysin inhibitor (ARNI) combination to replace ACE inhibitor or ARB if patient remains symptomatic<br><br>Hydralazine/isosorbide dinitrate combined for Black patients or if ACE inhibitor and ARB are contraindicated or not tolerated (see also <a href="#">Heart Failure</a> )<br><br>Thiazide or loop diuretic as additive therapy<br><br>Dihydropyridine CCB (except nifedipine) | Monitor serum K <sup>+</sup> and SCr with ARNI or the combination of ACE inhibitor, ARB or aldosterone antagonist. |
|                                                    | Left ventricular hypertrophy | ACE inhibitor, ARB, long-acting CCB or thiazide diuretic                                                                                        | Combinations of additional agents                                                                                                                                                                                                                                                                                                                                                                              | Hydralazine and minoxidil should not be used.                                                                      |
|                                                    | Past stroke or TIA           | ACE inhibitor/diuretic combination                                                                                                              | Combinations of additional agents                                                                                                                                                                                                                                                                                                                                                                              | Hypertension should not be treated in acute stroke unless BP extremely elevated.                                   |

| Category                                  | Risk Factor/Disease                                 | Initial Therapy                                                                                             | Second-Line Therapy               | Notes/Cautions                                                                                                                                                                                                                            |
|-------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           |                                                     |                                                                                                             |                                   | Combination of an ACE inhibitor with an ARB is specifically not recommended.                                                                                                                                                              |
| <b>Nondiabetic chronic kidney disease</b> | Nondiabetic chronic kidney disease with proteinuria | ACE inhibitor (ARB if ACE inhibitor not tolerated); diuretics as additive therapy                           | Combinations of additional agents | Carefully monitor serum K <sup>+</sup> and SCr in patients on an ACE inhibitor or an ARB.<br><br>Combination of an ACE inhibitor with an ARB is specifically not recommended in patients with chronic kidney disease without proteinuria. |
|                                           | Renovascular disease                                | Does not affect initial treatment recommendations                                                           | Combinations of additional agents | Avoid ACE inhibitors or ARBs in patients with bilateral renal artery stenosis or unilateral disease with a solitary kidney.                                                                                                               |
| <b>Other conditions</b>                   | Peripheral arterial disease                         | Does not affect initial treatment recommendations                                                           | Combinations of additional agents | Avoid beta-blockers in patients with severe disease.                                                                                                                                                                                      |
|                                           | Dyslipidemia                                        | Does not affect initial treatment recommendations                                                           | Combinations of additional agents |                                                                                                                                                                                                                                           |
|                                           | Overall vascular protection                         | Statin therapy for patients with hypertension and 3 or more CV risk factors or with atherosclerotic disease |                                   |                                                                                                                                                                                                                                           |

**Abbreviations:** ACE = angiotensin-converting enzyme; ARB = angiotensin II receptor blocker; BNP = brain natriuretic peptide; CCB = calcium channel blocker; CKD = chronic kidney disease; CV = cardiovascular; CVD = cardiovascular disease; DBP = diastolic blood pressure; MI = myocardial infarction; NT-proBNP = N-terminal-proBNP; NYHA = New York Heart Association; SBP = systolic blood pressure; SCr = serum creatinine; TIA = transient ischemic attack

## Choices during Pregnancy and Breastfeeding

### Hypertension and Pregnancy

Inform patients with pre-existing hypertension who are of childbearing potential, particularly those who are considering pregnancy, that they are at an increased risk of adverse pregnancy outcomes including intrauterine growth restriction; placental abruption; preterm delivery and the attendant neonatal risks of prematurity; and particularly a heightened risk of preeclampsia, with a crude risk of about 20–25% (varying with the severity and duration of the pre-existing hypertension). Enhanced surveillance is required during pregnancy (ideally by an obstetrician or perinatologist) to monitor for these complications. Prior to conception, or immediately upon recognition of an unplanned pregnancy, review the choice of antihypertensive medication and consider substitution of agents with the best safety profiles. Patients at high risk of preeclampsia (including all patients with chronic hypertension) should be offered **ASA** (81–162 mg taken daily at bedtime) beginning after conception, but prior to 16 weeks' gestational age and discontinued by around 36 weeks' gestation.<sup>[20][21][22]</sup> Patients with a low dietary intake of calcium (<900 mg/day) should receive supplementation of at least 500 mg daily of oral **calcium**.<sup>[20][23]</sup>

### Management

While there remains a dearth of high-quality data on the effects of many common antihypertensive medications on the developing fetus, international guidelines<sup>[24][25][26][27]</sup> have reached some consensus regarding a list of “preferred” medications for use in pregnancy, as well as a few “avoid” and “must avoid” drugs. Antihypertensive therapy during pregnancy should be initiated for sustained BP >140/90 and maintained or adjusted to achieve the BP target of 120–140/80–90. Medications widely considered first-line for the management of hypertension include **labetalol** (100 mg BID to 800 mg TID) and **nifedipine** XL (30 mg daily to 60 mg BID). **Methyldopa** (250 mg BID to 1000 mg TID) is also an acceptable choice, but is chosen less frequently due to its lesser potency and its frequent side effects.<sup>[25]</sup> These medications are preferred as they have evidence and/or a strong clinical record of safe and effective use in pregnancy<sup>[28][29]</sup> as well as an absence of demonstrated adverse effects on subsequent neonatal and childhood development. Other oral beta-blockers (**acebutolol**, **metoprolol**, **pindolol** and **propranolol**) can also be considered, though they are used infrequently for this indication. For the acute management of severe hypertension in pregnancy (SBP ≥160 mm Hg and/or DBP ≥110 mm Hg), rapid lowering of BP is recommended with either immediate-release oral **nifedipine** (5–10 mg every 30 min as required), parenteral labetalol (20 mg IV; repeat 20–80 mg IV every 30 min as required) or parenteral **hydralazine** (5 mg IV; repeat 5–10 mg IV every 30 minutes as required).<sup>[20]</sup> Other oral antihypertensive medications that can potentially be used in pregnancy after the above agents have been considered include **clonidine**, hydralazine and **thiazide diuretics**.<sup>[24]</sup>

Avoid atenolol, as its use for the treatment of hypertension in pregnancy has been associated with fetal intrauterine growth restriction (IUGR).<sup>[30]</sup> The other beta-blockers, in contrast, are only weakly associated with IUGR and have been used widely in pregnancy for various indications. Thiazide diuretics and **loop diuretics** may prevent the physiologic volume expansion seen in normal pregnancy, and thereby impair uteroplacental perfusion and fetal growth.<sup>[31]</sup> Available data do not support an adverse effect on perinatal outcome,<sup>[25]</sup> however, and these medications may therefore be considered or continued in patients felt to have volume-dependent hypertension (renal impairment). They should be avoided in settings in which uteroplacental perfusion is already reduced (preeclampsia or IUGR). Spironolactone should not be used at all in pregnancy, due to its anti-androgenic effects.<sup>[32]</sup>

ACE inhibitors have clearly been shown to be fetotoxic when taken during the second and third trimesters,<sup>[33]</sup> leading to oligohydramnios, IUGR, fetal/neonatal renal failure and other growth effects. First-trimester exposure has also been associated with teratogenic effects, mainly to the fetal CV and central nervous system, though these are likely due to confounding by undiagnosed diabetes.<sup>[34][35][36][37]</sup> These medications should generally be discontinued prior to conception or promptly upon discovery of an unplanned pregnancy. Cautious continuation during the periconceptual period may be considered in patients with proteinuric renal disease, along with frequent pregnancy testing and immediate switching to a preferred agent upon confirmation of pregnancy. The data regarding the risk of fetal harm from ARBs<sup>[38]</sup> and direct renin inhibitors<sup>[39]</sup> are less robust (mainly animal data), but they appear to have similar harmful effects and should be avoided just as strictly as ACE inhibitors during pregnancy.

Most patients with pre-existing hypertension, particularly those with long-standing, difficult-to-control hypertension or end-organ damage, should be followed throughout pregnancy by a specialist in obstetrics and gynecology. These clinicians are skilled at ongoing maternal management as well as appropriate monitoring of fetal growth and well-being. Patients with difficult-to-control hypertension or other medical issues benefit from assessment and follow-up with a hypertension specialist or obstetric medicine physician during their pregnancy.

### Hypertension and Breastfeeding

Following the completion of a pregnancy, many patients require ongoing antihypertensive therapy. The choice of antihypertensive agent may be influenced by whether or not the person is breastfeeding, as all oral medications appear in breast milk to some degree.<sup>[40]</sup> Those who are breastfeeding may safely continue treatment with any “pregnancy-preferred” drug in the postpartum period. These include labetalol, long-acting nifedipine and methyldopa. If the use of an ACE inhibitor is strongly favoured, preferred agents include captopril, enalapril, benazepril and quinapril.<sup>[41]</sup>

A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#), and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

## Therapeutic Tips

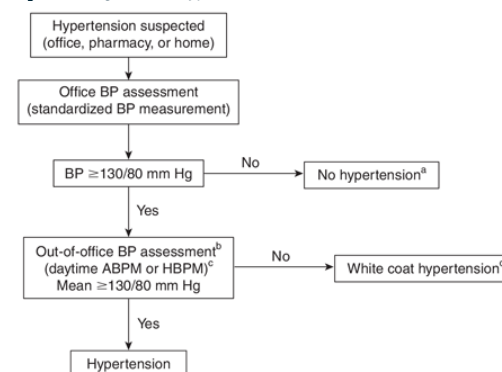
- Prescribe a lower starting dose of antihypertensive drugs in older patients.
- Recent onset of hypertension or change in BP control suggests an identifiable or secondary cause, such as drugs known to exacerbate hypertension or new onset of significant renal artery stenosis.
- Many drugs ineffective as monotherapy for controlling BP are effective components in a rational combination regimen.
- Consider concurrent CV risk factors and disease states when prescribing therapy (see [Table 5](#)).
- Cardiovascular risk can vary 10-fold in persons with the same BP. Assess global CV risk in all hypertensive patients using a risk form, chart or computer program (see Dyslipidemias, [Figure 1](#) as an example).
- Blood pressure readings provided in a pharmacy or taken at home should be considered only if taken correctly using a validated and Hypertension Canada–approved instrument. Blood pressure measurements taken at home correlate better with CV outcomes than office-based measurements.<sup>[42][43]</sup> A home BP series is the recommended method of taking home BP (see [Diagnosis](#)). Recommend monitors endorsed by Hypertension Canada and train patients to use the proper technique. To allay anxiety, caution patients that some variation throughout the day is normal. Patient instructions for selecting and using home BP monitors can be found in the public section of the [Hypertension Canada](#) website.
- Pharmacists and nurses can play an important role in hypertension screening, medication selection, patient education, follow-up and adherence monitoring. Dietitians can assist patients in managing their sodium and caloric intake.
- A team approach to hypertension management is more effective than usual care. In patients with hypertension and diabetes, joint care by a family physician, community pharmacist and nurse resulted in an approximately 6 mm Hg greater reduction in

SBP over 6 months, compared with usual physician-based care.<sup>[44]</sup>

- Combining home BP telemonitoring with pharmacist case management effectively reduces BP and improves control.<sup>[45]</sup>

## Algorithms

**Figure 1: Diagnosis of Hypertension**<sup>[6][7][8]</sup>



<sup>[a]</sup> Continue yearly or opportunistic BP monitoring. Perform out-of-office BP assessment if masked hypertension is suspected.

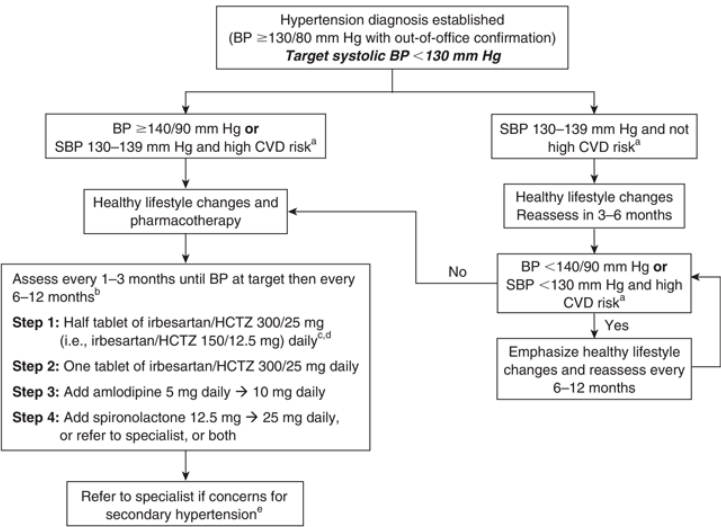
<sup>[b]</sup> If unavailable or infeasible, repeat office BP assessment at a separate visit.

<sup>[c]</sup> Use Hypertension Canada-recommended or other validated home BP devices.

<sup>[d]</sup> Continue yearly or opportunistic BP monitoring.

**Abbreviations:** ABPM = ambulatory blood pressure measurement; AOBP = automated office blood pressure; BP = blood pressure; HBPM = home blood pressure monitor

**Figure 2: Suggested Management of Hypertension**<sup>[6][7][8]</sup>



<sup>[a]</sup> High CVD risk conditions include established CVD (coronary artery disease, heart failure, cerebrovascular disease and peripheral artery disease), type 1 or type 2 diabetes mellitus, chronic kidney disease (eGFR <60 mL/min/1.73 m<sup>2</sup> or albuminuria ≥3 mg/mmol), 10-year Framingham Risk Score ≥20%, and age ≥75 years.

<sup>[b]</sup> Initiate pharmacotherapy at Step 1. If BP not at target when next assessed, move onto the next step. Progress through the steps until BP at target.

<sup>[c]</sup> Initial single-pill combination therapy is preferred if available and covered. If not, 2 separate pills can be used as an alternative, with preference for a long-acting thiazide-like diuretic over a thiazide diuretic. If patient is at high risk of symptomatic hypotension, treatment initiation with a single agent is reasonable. Angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers should be avoided or discontinued in all people who are pregnant or trying to become pregnant.

<sup>[d]</sup> See [Table 6](#) for acceptable alternative single-pill combinations available in Canada.

<sup>[e]</sup> Includes unexplained, sudden worsening of hypertension, hypokalemia, adrenal nodule, abdominal bruit, acute kidney injury after hypertension treatment, severe or malignant hypertension, hypertension at a young age, and resistant hypertension.

**Abbreviations:** BP = blood pressure; CVD = cardiovascular disease; eGFR = estimated glomerular filtration rate; HCTZ = hydrochlorothiazide; SBP = systolic blood pressure

Drug Table

Table 6: Drugs Used for Hypertension

| Drug/Cost <sup>[a]</sup>                               | Dosage                                     | Adverse Effects                                | Drug Interactions                            | Comments                 |
|--------------------------------------------------------|--------------------------------------------|------------------------------------------------|----------------------------------------------|--------------------------|
| Drug Class: Alpha <sub>1</sub> -Adrenergic Antagonists |                                            |                                                |                                              |                          |
| <a href="#">doxazosin</a> generics                     | Initial: 1 mg/day<br><br>Usual: 1–8 mg/day | Orthostatic hypotension, headache, drowsiness, | Caution when adding other hypotensive drugs, | Not for initial therapy. |
| <\$20                                                  |                                            |                                                |                                              |                          |

| Drug/Cost <sup>[a]</sup>                          | Dosage                                                                                                                               | Adverse Effects                                                                                                                                                                                                                                                    | Drug Interactions                                                                                                         | Comments                 |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------|
|                                                   | Maximum: 16 mg/day<br><br>Once daily PO                                                                                              | palpitations, nasal congestion. Syncope usually occurs at the start of therapy, with rapid dose titration or on addition of other agents. Titrate slowly. If interrupted for several days, restart at initial dose.                                                | as may cause syncope.                                                                                                     |                          |
| <a href="#">prazosin</a> generics<br><br><\$20–40 | Initial: 0.5 mg with p.m. meal (Day 1), then 0.5 mg BID–TID PO × 3 days and gradually increase as required<br><br>Maximum: 20 mg/day | Orthostatic hypotension, headache, drowsiness, palpitations, nasal congestion. Syncope usually occurs at the start of therapy, with rapid dose titration or on addition of other agents. Titrate slowly. If interrupted for several days, restart at initial dose. | Caution when adding other hypotensive drugs, as may cause syncope.                                                        | Not for initial therapy. |
| <a href="#">terazosin</a> generics<br><br><\$20   | Initial: 1 mg QHS PO<br><br>Usual: 1–5 mg/day<br><br>Maximum: 20 mg/day<br><br>Once daily or divided BID PO                          | Orthostatic hypotension, headache, drowsiness, palpitations, nasal congestion. Syncope usually occurs at the start of therapy, with rapid dose                                                                                                                     | Caution when adding other hypotensive drugs, as may cause syncope. Verapamil increases serum concentrations of terazosin. | Not for initial therapy. |



| Drug/Cost <sup>[a]</sup>                                   | Dosage                       | Adverse Effects                                                                                                                                                    | Drug Interactions                                                                                                                                                                                                                                                                                                     | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                            |                              | titration or on addition of other agents. Titrate slowly. If interrupted for several days, restart at initial dose.                                                |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Drug Class: Angiotensin-Converting Enzyme (ACE) Inhibitors |                              |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>benazepril</b> ⓘ<br>generics<br><br>\$20–40             | Initial: 10 mg/day           | Dry cough, hyperkalemia, angioedema (unusual). Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> . | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly. |
|                                                            | Usual: 20 mg/day             |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                            | Maximum: 40 mg/day           |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                            | Once daily or divided BID PO |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                            |                              |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>captopril</b> ⓘ<br>generics<br><br>\$20–40              | Initial: 25 mg/day           | Dry cough, hyperkalemia, angioedema (unusual). Can precipitate renal failure in renovascular disease, volume                                                       | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect                                                                                                                                                               | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup>                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                            | Usual: 75 mg/day             |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                            | Maximum: 150 mg/day          |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                            | Divided BID or TID PO        |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                            |                              |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Drug/Cost <sup>[a]</sup>                               | Dosage                       | Adverse Effects                                                                                                                                                    | Drug Interactions                                                                                                                                                                                                                                                                                                     | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                        |                              | depletion or those receiving <a href="#">NSAIDs</a> .                                                                                                              | with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose).                                                                                                                                                         | Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly.                                                                                                      |
|                                                        |                              |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>cilazapril</b> ⓘ<br>Inhibace, generics<br><br><\$20 | Initial: 2.5 mg/day          | Dry cough, hyperkalemia, angioedema (unusual). Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> . | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few |
|                                                        | Usual: 2.5–5 mg/day          |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                        | Maximum: 10 mg/day           |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                        | Once daily or divided BID PO |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                        |                              |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Drug/Cost <sup>[a]</sup>                                                                                                             | Dosage                                                                                                     | Adverse Effects                                                                                                                                                       | Drug Interactions                                                                                                                                                                                                                                                                                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                      |                                                                                                            |                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                          | days, then regularly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>enalapril</b> <br>Vasotec, generics<br><br><\$20 | Initial: 5 mg/day<br><br>Usual: 10–40 mg/day<br><br>Maximum: 40 mg/day<br><br>Once daily or divided BID PO | Dry cough, hyperkalemia, angioedema (unusual).<br>Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> . | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly. |
| <b>fosinopril</b><br>generics<br><br><\$20                                                                                           | Initial: 10 mg/day<br><br>Usual: 20 mg/day<br><br>Maximum: 40 mg/day<br><br>Once daily or divided BID PO   | Dry cough, hyperkalemia, angioedema (unusual).<br>Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> . | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br>Hyperkalemia usually occurs only in those on K <sup>+</sup>                                                                                                                                                                                                                            |

| Drug/Cost <sup>[a]</sup>                                                                                                                | Dosage                                                                                    | Adverse Effects                                                                                                                                                       | Drug Interactions                                                                                                                                                                                                                                                                                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                         |                                                                                           |                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                          | supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly.                                                                                                                                                                                                                                                                                           |
| <b>lisinopril</b> <br>Zestril, generics<br><br><\$20 | Initial: 10 mg/day<br><br>Usual: 20 mg/day<br><br>Maximum: 40 mg/day<br><br>Once daily PO | Dry cough, hyperkalemia, angioedema (unusual).<br>Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> . | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly. |

| Drug/Cost <sup>[a]</sup>                                                                                                                         | Dosage                                                 | Adverse Effects                                                                                                                                                    | Drug Interactions                                                                                                                                                                                                                                                                                                     | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">perindopril</a> <br>Coversyl, generics<br><br><\$20 | Initial: 4 mg/day                                      | Dry cough, hyperkalemia, angioedema (unusual). Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> . | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly. |
|                                                                                                                                                  | Maximum: 8 mg/day<br><br>Once daily or divided BID PO  |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <a href="#">quinapril</a> <br>generics<br><br><\$20             | Initial: 10 mg/day                                     | Dry cough, hyperkalemia, angioedema (unusual). Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> . | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal                                                                                                                                                 |
|                                                                                                                                                  | Maximum: 40 mg/day<br><br>Once daily or divided BID PO |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Drug/Cost <sup>[a]</sup>                                                                                                                                            | Dosage                                                                         | Adverse Effects                                                                                                                                                    | Drug Interactions                                                                                                                                                                                                                                                                                                     | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">ramipril</a> <br>Altace, generics<br><br><\$20                       | Initial: 2.5 mg/day                                                            | Dry cough, hyperkalemia, angioedema (unusual). Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> . | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly. |
|                                                                                                                                                                     | Usual: 10 mg/day<br><br>Maximum: 20 mg/day<br><br>Once daily or divided BID PO |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <a href="#">trandolapril</a> <br><a href="#">Mavik</a> , generics<br><br><\$20 | Initial: 1 mg/day                                                              | Dry cough, hyperkalemia, angioedema (unusual). Can precipitate renal failure in renovascular disease, volume depletion or                                          | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and                                                                                                                               | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br>Use lower (50%) initial doses if on diuretics (increased                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                                                                                     | Maximum: 4 mg/day<br><br>Once daily PO                                         |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Drug/Cost <sup>[a]</sup>                         | Dosage | Adverse Effects                          | Drug Interactions                                                                                                             | Comments                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------|--------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  |        | those receiving <a href="#">NSAIDs</a> . | increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | risk of hypotension with hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly. |
| Drug Class: Angiotensin Receptor Blockers (ARBs) |        |                                          |                                                                                                                               |                                                                                                                                                                                                                                                                                                                                      |

| Drug/Cost <sup>[a]</sup>                                      | Dosage                                                            | Adverse Effects                                                                                                                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <a href="#">azilsartan</a><br>Edarbi, generics<br><br>\$20–40 | Initial: 40 mg/day<br><br>Maximum: 80 mg/day<br><br>Once daily PO | Hyperkalemia.<br>Can precipitate renal failure in susceptible patients (bilateral renovascular disease, those with volume depletion or with concurrent <a href="#">NSAID</a> use).<br><br>Angioedema has been reported, but a causal association has not been established. | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[14]</sup><br>Use lower initial doses in patients who are volume depleted or on diuretics (increased risk of hypotension in hypovolemia).<br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or diabetics with high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few days, then regularly. |
|                                                               |                                                                   |                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Drug/Cost <sup>[a]</sup>                                     | Dosage                                                           | Adverse Effects                                                                                                                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>candesartan</b><br><i>Atacand</i> , generics<br><br><\$20 | Initial: 8 mg/day<br><br>Usual: 8–16 mg/day<br><br>Once daily PO | Hyperkalemia.<br>Can precipitate renal failure in susceptible patients (bilateral renovascular disease, those with volume depletion or with concurrent <a href="#">NSAID</a> use).<br><br>Angioedema has been reported, but a causal association has not been established. | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br><br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[14]</sup><br><br>Use lower initial doses in patients who are volume depleted or on diuretics (increased risk of hypotension in hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or diabetics with high serum K <sup>+</sup> levels.<br>Assess SCr and K <sup>+</sup> after a few days, then regularly. |

| Drug/Cost <sup>[a]</sup>                                   | Dosage                                                                | Adverse Effects                                                                                                                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>irbesartan</b><br><i>Avapro</i> , generics<br><br><\$20 | Initial: 150 mg/day<br><br>Usual: 150–300 mg/day<br><br>Once daily PO | Hyperkalemia.<br>Can precipitate renal failure in susceptible patients (bilateral renovascular disease, those with volume depletion or with concurrent <a href="#">NSAID</a> use).<br><br>Angioedema has been reported, but a causal association has not been established. | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br><br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[14]</sup><br><br>Use lower initial doses in patients who are volume depleted or on diuretics (increased risk of hypotension in hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or diabetics with high serum K <sup>+</sup> levels.<br>Assess SCr and K <sup>+</sup> after a few days, then regularly. |

| Drug/Cost <sup>[a]</sup>                                  | Dosage                                                                                                           | Adverse Effects                                                                                                                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <a href="#">losartan</a><br>Cozaar, generics<br><br><\$20 | Initial: 50 mg/day<br><br>Usual: 25–100 mg/day<br><br>Maximum: 100 mg/day<br><br>Once daily or divided BID<br>PO | Hyperkalemia.<br>Can precipitate renal failure in susceptible patients (bilateral renovascular disease, those with volume depletion or with concurrent <a href="#">NSAID</a> use).<br><br>Angioedema has been reported, but a causal association has not been established. | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br><br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[14]</sup><br><br>Use lower initial doses in patients who are volume depleted or on diuretics (increased risk of hypotension in hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or diabetics with high serum K <sup>+</sup> levels.<br>Assess SCr and K <sup>+</sup> after a few days, then regularly. |

| Drug/Cost <sup>[a]</sup>                                     | Dosage                                                            | Adverse Effects                                                                                                                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                            | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <a href="#">olmesartan</a><br>Olmetec, generics<br><br><\$20 | Initial: 20 mg/day<br><br>Maximum: 40 mg/day<br><br>Once daily PO | Hyperkalemia.<br>Can precipitate renal failure in susceptible patients (bilateral renovascular disease, those with volume depletion or with concurrent <a href="#">NSAID</a> use).<br><br>Angioedema has been reported, but a causal association has not been established. | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br><br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[14]</sup><br><br>Use lower initial doses in patients who are volume depleted or on diuretics (increased risk of hypotension in hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or diabetics with high serum K <sup>+</sup> levels.<br>Assess SCr and K <sup>+</sup> after a few days, then regularly. |

| Drug/Cost <sup>[a]</sup>                                               | Dosage                                                          | Adverse Effects                                                                                                                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>telmisartan</b><br><a href="#">Micardis</a> , generics<br><br><\$20 | Initial: 80 mg/day<br><br>Usual: 80 mg/day<br><br>Once daily PO | Hyperkalemia.<br>Can precipitate renal failure in susceptible patients (bilateral renovascular disease, those with volume depletion or with concurrent <a href="#">NSAID</a> use).<br><br>Angioedema has been reported, but a causal association has not been established. | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[14]</sup><br><br>Use lower initial doses in patients who are volume depleted or on diuretics (increased risk of hypotension in hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or diabetics with high serum K <sup>+</sup> levels.<br>Assess SCr and K <sup>+</sup> after a few days, then regularly. |

| Drug/Cost <sup>[a]</sup>                                           | Dosage                                                              | Adverse Effects                                                                                                                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                        | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <b>valsartan</b><br><a href="#">Diovan</a> , generics<br><br><\$20 | Initial: 80 mg/day<br><br>Usual: 80–320 mg/day<br><br>Once daily PO | Hyperkalemia.<br>Can precipitate renal failure in susceptible patients (bilateral renovascular disease, those with volume depletion or with concurrent <a href="#">NSAID</a> use).<br><br>Angioedema has been reported, but a causal association has not been established. | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose). | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[14]</sup><br><br>Use lower initial doses in patients who are volume depleted or on diuretics (increased risk of hypotension in hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or diabetics with high serum K <sup>+</sup> levels.<br>Assess SCr and K <sup>+</sup> after a few days, then regularly. |

Drug Class: Beta<sub>1</sub>-Adrenergic Antagonists, nonselective

|                                                                                                                                 |                                                                                             |                                                                                                                                                              |                                                                                                                                                                |                                                                                                                                                                                                                                                                                          |
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| <b>nadolol</b> <br>generics<br><br>\$20–40 | Initial: 20 mg/day<br><br>Usual: 160 mg/day<br><br>Maximum: 320 mg/day<br><br>Once daily PO | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams.<br>Less common: hyperglycemia, depression, heart failure, heart block. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone. | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated.<br>Avoid in patients with asthma. <sup>[48]</sup><br><br>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation. |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Drug/Cost <sup>t</sup> <sup>a</sup>                         | Dosage                                                                                                                      | Adverse Effects                                                                                                                                              | Drug Interactions                                                                                                                                                                                                                                                                 | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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|                                                             |                                                                                                                             |                                                                                                                                                              |                                                                                                                                                                                                                                                                                   | Avoid in patients with severe <a href="#">PAD</a> .<br>Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker.                                                                                                                                                                                                                                                                                                 |
| <a href="#">propranolol, controlled-release</a><br>generics | Initial: 80 mg/day<br><br>Usual: 320 mg/day<br><br>Maximum: 480 mg/day<br><br>SR (once daily PO)<br>formulation recommended | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams.<br>Less common: hyperglycemia, depression, heart failure, heart block. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone.<br><br>CYP2D6 inhibitors increase levels of propranolol.<br><br>Propranolol increases serum levels of rizatriptan. | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated.<br>Avoid in patients with asthma. <sup>[46]</sup><br><br>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.<br><br>Avoid in patients with severe <a href="#">PAD</a> .<br>Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker. |
| \$60–80                                                     |                                                                                                                             |                                                                                                                                                              |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <a href="#">timolol</a><br>generics                         | Initial: 5 mg BID                                                                                                           | Fatigue, bradycardia,                                                                                                                                        | Bradycardia with digoxin or                                                                                                                                                                                                                                                       | Beta-blockers should not be used                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Drug/Cost <sup>t</sup> <sup>a</sup>                                                                                                | Dosage                                                                                                    | Adverse Effects                                                                                                                                                                                                        | Drug Interactions                                                                                                                                              | Comments                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                                                                                                                                    | Usual: 20 mg BID<br><br>Maximum: 30 mg BID PO                                                             | decreased exercise capacity, headache, impotence, vivid dreams.<br>Less common: hyperglycemia, depression, heart failure, heart block.                                                                                 | nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone.                             | as initial therapy in patients >60 y of age unless specifically indicated.<br>Avoid in patients with asthma. <sup>[46]</sup><br><br>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.<br><br>Avoid in patients with severe <a href="#">PAD</a> .<br>Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker. |
| \$20–40                                                                                                                            |                                                                                                           |                                                                                                                                                                                                                        |                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Drug Class: Beta <sub>1</sub> -Adrenergic Antagonists, beta <sub>1</sub> -selective                                                |                                                                                                           |                                                                                                                                                                                                                        |                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <a href="#">atenolol</a> <br>Tenormin, generics | Initial: 25 mg/day<br><br>Usual: 50 mg/day<br><br>Maximum: 100 mg/day<br><br>Once daily or divided BID PO | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams.<br>Less common: hyperglycemia, depression, heart failure, heart block.<br><br>Fewer noncardiac effects due to cardioselectivity. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone. | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated.<br>Avoid in patients with asthma. <sup>[46]</sup><br><br>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.<br><br>Avoid in patients with severe <a href="#">PAD</a> .<br>Contraindicated in patients with second- or third-                 |
| <\$20                                                                                                                              |                                                                                                           |                                                                                                                                                                                                                        |                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                           |



| Drug/Cost <sup>[a]</sup>                                                                                                     | Dosage                                                                                                                                            | Adverse Effects                                                                                                                                                                                                        | Drug Interactions                                                                                                                                                                                                      | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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|                                                                                                                              |                                                                                                                                                   |                                                                                                                                                                                                                        |                                                                                                                                                                                                                        | degree heart block in the absence of a pacemaker.                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>bisoprolol</b> <br>generics<br><br><\$20 | Initial: 5 mg/day<br><br>Usual: 10 mg/day<br><br>Maximum: 20 mg/day<br><br>Once daily PO                                                          | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams.<br>Less common: hyperglycemia, depression, heart failure, heart block.<br><br>Fewer noncardiac effects due to cardioselectivity. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone.                                                         | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated.<br>Avoid in patients with asthma. <sup>[46]</sup><br><br>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.<br><br>Avoid in patients with severe <a href="#">PAD</a> .<br><br>Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker. |
| <b>metoprolol</b><br>generics<br><br><\$20                                                                                   | Initial: 50 mg/day<br><br>Usual: 100–200 mg/day<br><br>Maximum: 400 mg/day<br><br>Give regular formulations BID PO; SR formulations once daily PO | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams.<br>Less common: hyperglycemia, depression, heart failure, heart block.<br><br>Fewer noncardiac                                   | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone.<br><br>CYP2D6 inhibitors increase levels of metoprolol. | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated.<br>Avoid in patients with asthma. <sup>[46]</sup><br><br>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.                                                                                                                                                                       |

| Drug/Cost <sup>[a]</sup>                                                                                                        | Dosage                                                                                   | Adverse Effects                                                                                                                                                                                                        | Drug Interactions                                                                                                                                                                                                     | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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|                                                                                                                                 |                                                                                          | effects due to cardioselectivity.                                                                                                                                                                                      |                                                                                                                                                                                                                       | Avoid in patients with severe <a href="#">PAD</a> .<br><br>Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker.                                                                                                                                                                                                                                                                                                 |
| <b>nebivolol</b> <br>Bystolic<br><br>\$40–60 | Initial: 5 mg/day<br><br>Usual: 10 mg/day<br><br>Maximum: 20 mg/day<br><br>Once daily PO | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams.<br>Less common: hyperglycemia, depression, heart failure, heart block.<br><br>Fewer noncardiac effects due to cardioselectivity. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone.<br><br>CYP2D6 inhibitors increase levels of nebivolol. | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated.<br>Avoid in patients with asthma. <sup>[46]</sup><br><br>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.<br><br>Avoid in patients with severe <a href="#">PAD</a> .<br><br>Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker. |
| Drug Class: Beta <sub>1</sub> -Adrenergic Antagonists, nonselective with intrinsic sympathomimetic activity (ISA)               |                                                                                          |                                                                                                                                                                                                                        |                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>pindolol</b><br>Visken, generics<br><br>\$40–60                                                                              | Initial: 5 mg BID PO<br><br>Usual: 15 mg BID PO<br><br>Maximum: 60 mg/day                | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams.<br>Less common: hyperglycemia,                                                                                                   | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone.                                                        | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated.<br>Avoid in patients with asthma. <sup>[46]</sup>                                                                                                                                                                                                                                                                                                  |

| Drug/Cost <sup>[a]</sup> | Dosage | Adverse Effects                         | Drug Interactions | Comments                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|--------|-----------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |        | depression, heart failure, heart block. |                   | <p>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.</p> <p>Avoid in patients with severe <a href="#">PAD</a>.</p> <p>Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker.</p> <p>Agents with ISA have less effect on resting heart rate than those without ISA.</p> |

| Drug Class: Beta <sub>1</sub> -Adrenergic Antagonists, beta <sub>1</sub> -selective with ISA                             |                                                                                                             |                                                                                                                                                                                                                     |                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <a href="#">acebutolol</a> <br>generics | Initial: 100 mg/day<br><br>Usual: 400 mg/day<br><br>Maximum: 800 mg/day<br><br>Once daily or divided BID PO | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams. Less common: hyperglycemia, depression, heart failure, heart block.<br><br>Fewer noncardiac effects due to cardioselectivity. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> . Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone. | <p>Beta-blockers should not be used as initial therapy in patients &gt;60 y of age unless specifically indicated.</p> <p>Avoid in patients with asthma.<sup>[46]</sup></p> <p>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.</p> <p>Avoid in patients with severe <a href="#">PAD</a>.</p> <p>Contraindicated in patients with second- or third-degree heart block</p> |

| Drug/Cost <sup>[a]</sup>                                                                         | Dosage                                                                        | Adverse Effects                                                                                                                                           | Drug Interactions                                                                                                                                           | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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|                                                                                                  |                                                                               |                                                                                                                                                           |                                                                                                                                                             | <p>in the absence of a pacemaker.</p> <p>Agents with <a href="#">ISA</a> have less effect on resting heart rate than those without <a href="#">ISA</a>.</p>                                                                                                                                                                                                                                                                                                                |
| Drug Class: Beta <sub>1</sub> -Adrenergic Antagonists with alpha <sub>1</sub> -blocking activity |                                                                               |                                                                                                                                                           |                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <a href="#">labetalol</a><br>generics                                                            | Initial: 50 mg BID PO<br><br>Usual: 200 mg BID PO<br><br>Maximum: 1200 mg/day | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams. Less common: hyperglycemia, depression, heart failure, heart block. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> . Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone. | <p>Beta-blockers should not be used as initial therapy in patients &gt;60 y of age unless specifically indicated.</p> <p>Avoid in patients with asthma.<sup>[46]</sup></p> <p>Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation.</p> <p>Avoid in patients with severe <a href="#">PAD</a>.</p> <p>Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker.</p> |
| \$20–40                                                                                          |                                                                               | Edema, dizziness, nasal congestion and postural hypotension due to alpha <sub>1</sub> antagonism.                                                         |                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Drug Class: Calcium Channel Blockers, dihydropyridine            |                                                                    |                                                |                                                                                                                                          |  |
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| <a href="#">amlodipine</a><br><a href="#">Norvasc</a> , generics | Initial: 2.5 mg/day<br><br>Maximum: 10 mg/day<br><br>Once daily PO | Ankle edema, flushing, headache, palpitations. | CYP3A4 substrate (many potential interactions). Strong inhibitors include azole antifungals, protease inhibitors, macrolides, quinidine. |  |
| <\$20                                                            |                                                                    |                                                |                                                                                                                                          |  |

| Drug/Cost <sup>[a]</sup>                                                         | Dosage                                                                                                                                            | Adverse Effects                                                                         | Drug Interactions                                                                                                                                                                                      | Comments                                                                                                        |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|                                                                                  |                                                                                                                                                   |                                                                                         | Grapefruit juice may increase serum concentrations.                                                                                                                                                    |                                                                                                                 |
| <b>felodipine, extended-release</b><br>Plendil, generics<br><\$20                | Initial: 2.5 mg/day<br>Usual: 10 mg/day<br>Maximum: 20 mg/day<br>Once daily PO                                                                    | Ankle edema, flushing, headache, palpitations.                                          | CYP3A4 substrate (many potential interactions).<br>Strong inhibitors include azole antifungals, protease inhibitors, macrolides, quinidine.<br><br>Grapefruit juice may increase serum concentrations. | Grapefruit juice causes marked elevations in felodipine serum levels and adverse events.                        |
| <b>nifedipine, extended-release</b><br>Adalat XL, Mylan-Nifedipine ER<br>\$20–40 | Initial: 30 mg/day<br>Usual: 60 mg/day<br>Maximum: 120 mg/day<br>Once daily PO                                                                    | Ankle edema, flushing, headache, palpitations.                                          | CYP3A4 substrate (many potential interactions).<br>Strong inhibitors include azole antifungals, protease inhibitors, macrolides, quinidine.<br><br>Grapefruit juice may increase serum concentrations. | Do not use short-acting nifedipine formulations for treatment of essential hypertension.                        |
| Drug Class: Calcium Channel Blockers, nondihydropyridine                         |                                                                                                                                                   |                                                                                         |                                                                                                                                                                                                        |                                                                                                                 |
| <b>diltiazem</b><br>Tiazac, Tiazac XC, generics<br><\$20                         | Initial: 120 mg/day<br>Usual: 240–360 mg/day<br>Maximum: 360 mg/day<br><br>Give CD or XC formulation once daily PO, SR formulation divided BID PO | Headache, dizziness, bradycardia, heart block, new onset or worsening of heart failure. | CYP3A4 substrate (many potential interactions).<br>Strong inhibitors include azole antifungals, protease inhibitors, macrolides, quinidine.<br><br>Grapefruit juice may increase serum concentrations. | Caution in patients with heart failure, or second- or third-degree heart block without a functioning pacemaker. |

| Drug/Cost <sup>[a]</sup>                             | Dosage                                                                                                                                                    | Adverse Effects                                                                                       | Drug Interactions                                                                                                                                                                                                                                                                                                                                                                                                                                            | Comments                                                                                                        |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|                                                      |                                                                                                                                                           |                                                                                                       | Nondihydropyridines inhibit the metabolism of carbamazepine, cyclosporine, lovastatin, simvastatin. Rifampin induces metabolism of nondihydropyridines.<br><br>Additive negative chronotropic effects with amiodarone, beta-blockers, digoxin.                                                                                                                                                                                                               |                                                                                                                 |
| <b>verapamil</b><br>Isoptin SR, generics<br><\$20–40 | Initial: 80 mg TID PO<br>Maximum: 160 mg TID PO<br><br>SR (once daily or divided BID PO): initial: 180 mg/day; usual: 180–480 mg/day; maximum: 480 mg/day | Headache, dizziness, bradycardia, heart block, new onset or worsening of heart failure. Constipation. | CYP3A4 substrate (many potential interactions).<br>Strong inhibitors include azole antifungals, protease inhibitors, macrolides, quinidine.<br><br>Grapefruit juice may increase serum concentrations.<br><br>Nondihydropyridines inhibit the metabolism of carbamazepine, cyclosporine, lovastatin, simvastatin. Rifampin induces metabolism of nondihydropyridines.<br><br>Additive negative chronotropic effects with amiodarone, beta-blockers, digoxin. | Caution in patients with heart failure, or second- or third-degree heart block without a functioning pacemaker. |

| Drug/Cost <sup>1a</sup>                                                                                                        | Dosage                                                                                                 | Adverse Effects                                                                                                                             | Drug Interactions                                                                                                                                                                           | Comments                                                                                                                                                                  |
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|                                                                                                                                |                                                                                                        |                                                                                                                                             | Verapamil increases digoxin levels by 50–75% within 1 wk (monitor levels).                                                                                                                  |                                                                                                                                                                           |
| Drug Class: Centrally Acting Antihypertensive Agents                                                                           |                                                                                                        |                                                                                                                                             |                                                                                                                                                                                             |                                                                                                                                                                           |
| <b>clonidine</b><br>generics<br><br><\$20                                                                                      | Initial: 0.1 mg BID PO<br><br>Usual: 0.3 mg BID PO                                                     | Sedation, dry mouth, orthostatic hypotension, dizziness.                                                                                    | Potential for bradycardia and AV block with beta-blockers, <a href="#">CCBs</a> and digoxin. Tricyclic antidepressants may reduce hypotensive effect.                                       | Not for initial therapy. Do not stop treatment abruptly. Dose must be reduced over at least 2–4 days to avoid withdrawal effects.                                         |
| <b>methyldopa</b> <br>generics<br><br>\$20–40 | Initial: 500 mg/day<br><br>Usual: 2000 mg/day<br><br>Maximum: 3000 mg/day<br><br>Divided BID or TID PO | Drowsiness, dry mouth, nasal congestion, depression, orthostatic hypotension, palpitations, sexual dysfunction, sodium and water retention. | Iron salts reduce absorption (separate administration). Additive hypotension with levodopa.<br><br>May exacerbate Li <sup>+</sup> adverse events without increasing Li <sup>+</sup> levels. | Positive Coombs test is common, but usually unimportant; hemolytic anemia is rare. Drug fever with or without an influenza-like illness; hepatic disorders have occurred. |
| Drug Class: Diuretics, potassium-sparing                                                                                       |                                                                                                        |                                                                                                                                             |                                                                                                                                                                                             |                                                                                                                                                                           |
| <b>amiloride</b> <br>Midamor<br><br><\$20   | Initial: 5 mg/day<br><br>Usual: 10 mg/day<br><br>Once daily PO                                         | Hyperkalemia, muscle cramps, headaches, GI symptoms.                                                                                        | Avoid <a href="#">NSAIDs</a> , <a href="#">ACE</a> inhibitors, aliskiren, angiotensin II receptor blockers, K <sup>+</sup> supplements: may cause severe hyperkalemia.                      | Not for initial therapy. Caution in significant renal function impairment, hyperkalemia.                                                                                  |

| Drug/Cost <sup>1a</sup>                                                                                                                       | Dosage                                                                 | Adverse Effects                                                                                                                                                                                                                                                                    | Drug Interactions                                                                                                                                                                                                                                                     | Comments                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>spironolactone</b> <br>Aldactone, generics<br><br><\$20 | Initial: 25 mg/day<br><br>Usual: 100 mg/day<br><br>Once daily PO       | Hyperkalemia, gynecomastia, androgen-like side effects, breast tenderness, GI symptoms.                                                                                                                                                                                            | Avoid <a href="#">NSAIDs</a> , <a href="#">ACE</a> inhibitors, aliskiren, angiotensin II receptor blockers, K <sup>+</sup> supplements: may cause severe hyperkalemia.                                                                                                | Not for initial therapy. Caution in significant renal function impairment, hyperkalemia.                                                                                                                                                                                                                                    |
| Drug Class: Diuretics, thiazide                                                                                                               |                                                                        |                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                             |
| <b>hydrochlorothiazide</b> <br>generics<br><br><\$20       | Initial: 12.5 mg/day<br><br>Usual: 25 mg/day<br><br>Once daily PO      | Hypotension, weakness, muscle cramps, impotence. Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.<br><br>Rare: azotemia, blood dyscrasias, allergic reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity. | Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose). <a href="#">NSAIDs</a> reduce hypotensive efficacy.<br><br>Diuretic-induced hypokalemia increases the risk of digoxin toxicity.<br><br>Reduced efficacy of antihyperglycemic agents. | Particularly effective in ISH, older and Black patients. Monitor SCr and K <sup>+</sup> .<br><br>Consider alternatives in patients with or predisposed to arrhythmias.<br><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low doses). Ineffective in patients with ClCr <30–40 mL/min. |
| <b>chlorthalidone</b> <br>generics<br><br><\$20          | Initial: 12.5 mg/day<br><br>Usual: 12.5–25 mg/day<br><br>Once daily PO | Hypotension, weakness, muscle cramps, impotence. Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.<br><br>Rare: azotemia, blood dyscrasias, allergic                                                                                                        | Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose). <a href="#">NSAIDs</a> reduce hypotensive efficacy.<br><br>Diuretic-induced hypokalemia increases the risk of digoxin toxicity.<br><br>Reduced efficacy of antihyperglycemic         | Lowest available tablet strength is 50 mg. Tablet (or “pill”) splitters, widely available in pharmacies, can be used to derive a dose of 12.5 mg (one-quarter tablet) with reasonable accuracy.<br><br>Particularly effective in ISH,                                                                                       |

| Drug/Cost <sup>[a]</sup> | Dosage | Adverse Effects                                                                                        | Drug Interactions | Comments                                                                                                                                                                                                                                                                                            |
|--------------------------|--------|--------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |        | reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity. | agents.           | older and Black patients. Monitor SCr and K <sup>+</sup> .<br><br>Consider alternatives in patients with or predisposed to arrhythmias.<br><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low doses).<br><br>Ineffective in patients with ClCr <30–40 mL/min. |

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| <b>indapamide</b> <br>generics | Initial: 1.25 mg/day<br><br>Usual: 2.5 mg/day<br><br>Once daily PO | Hypotension, weakness, muscle cramps, impotence. Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.<br><br>Rare: azotemia, blood dyscrasias, allergic reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity. | Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose). <a href="#">NSAIDs</a> reduce hypotensive efficacy.<br><br>Diuretic-induced hypokalemia increases the risk of digoxin toxicity.<br><br>Reduced efficacy of antihyperglycemic agents. | Particularly effective in ISH, older and Black patients. Monitor SCr and K <sup>+</sup> .<br><br>Consider alternatives in patients with or predisposed to arrhythmias.<br><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low doses). |
| <\$20                                                                                                           |                                                                    |                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                            |

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| <b>metolazone</b> <br>Zaroxolyn | Initial: 2.5 mg/day<br><br>Usual: 5 mg/day<br><br>Maximum: 10 mg/day | Hypotension, weakness, muscle cramps, impotence. | Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose). | Particularly effective in ISH, older and Black patients. |
| <\$20                                                                                                              |                                                                      |                                                  |                                                                                  |                                                          |

| Drug/Cost <sup>[a]</sup> | Dosage        | Adverse Effects                                                                                                                                                                                                                   | Drug Interactions                                                                                                                                                                    | Comments                                                                                                                                                                                                                                                                                                |
|--------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | Once daily PO | Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.<br><br>Rare: azotemia, blood dyscrasias, allergic reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity. | <a href="#">NSAIDs</a> reduce hypotensive efficacy.<br><br>Diuretic-induced hypokalemia increases the risk of digoxin toxicity.<br><br>Reduced efficacy of antihyperglycemic agents. | Monitor SCr and K <sup>+</sup> .<br><br>Consider alternatives in patients with or predisposed to arrhythmias.<br><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low doses).<br><br>Metolazone is effective in patients with moderate to severe renal dysfunction. |

|                                                                                                                            |                                                                                        |                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drug Class: ACE Inhibitor/Calcium Channel Blocker Combinations                                                             |                                                                                        |                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>perindopril</b> <br>Viacoram, generics | Initial: 3.5/2.5 mg<br><br>Usual: 7/5 mg<br><br>Maximum: 14/10 mg<br><br>Once daily PO | Dry cough, hyperkalemia, angioedema (unusual). Can precipitate renal failure in renovascular disease, volume depletion or those receiving <a href="#">NSAIDs</a> .<br><br>Ankle edema, flushing, headache, palpitations. | Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose).<br><br>CYP3A4 substrate (many potential interactions). Strong inhibitors include azole antifungals, protease inhibitors, macrolides, quinidine. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[12][13]</sup><br><br>Use lower (50%) initial doses if on diuretics (increased risk of hypotension with hypovolemia).<br><br>Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or drugs that cause K <sup>+</sup> retention, those with renal impairment or patients with diabetes who have high serum K <sup>+</sup> levels. Assess SCr and K <sup>+</sup> after a few |
| \$20–40                                                                                                                    |                                                                                        |                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| Drug/Cost <sup>1a</sup>                                                                                                                                                                                                                                    | Dosage                              | Adverse Effects                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                                                              | Comments                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                            |                                     |                                                                                                                                                            | Grapefruit juice may increase serum concentrations.                                                                                                                                                                                                                                                                                                            | days, then regularly.                                                                                                                                                                                                                                                                       |
| Drug Class: ACE Inhibitor/Diuretic Combinations                                                                                                                                                                                                            |                                     |                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                             |
| <b>cilazapril</b>  / <b>hydrochlorothiazide</b> <br>Inhibace Plus, generics<br><br><\$20 | 5/12.5 mg once daily PO             | Combination of <a href="#">ACE</a> inhibitor and diuretic adverse effects possible, e.g., dry cough, electrolyte disturbances, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose). Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses). Monitor SCr and K <sup>+</sup> . Ineffective in patients with ClCr <30–40 mL/min. |
| <b>enalapril</b>  / <b>hydrochlorothiazide</b> <br>Vaseretic, generics<br><br>\$20–40      | 5/12.5 mg or 10/25 mg once daily PO | Combination of <a href="#">ACE</a> inhibitor and diuretic adverse effects possible, e.g., dry cough, electrolyte disturbances, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose). Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses). Monitor SCr and K <sup>+</sup> . Ineffective in patients with ClCr <30–40 mL/min. |

| Drug/Cost <sup>1a</sup>                                                                                                                                                                                                                                                                    | Dosage                                           | Adverse Effects                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                                                              | Comments                                                                                                                                                                                                                                                                                    |
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| <b>lisinopril</b>  / <b>hydrochlorothiazide</b> <br>Zestoretic, generics<br><br><\$20                                | 10/12.5 mg, 20/12.5 mg or 20/25 mg once daily PO | Combination of <a href="#">ACE</a> inhibitor and diuretic adverse effects possible, e.g., dry cough, electrolyte disturbances, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose). Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses). Monitor SCr and K <sup>+</sup> . Ineffective in patients with ClCr <30–40 mL/min. |
| <b>perindopril</b>  / <b>indapamide</b> <br>Coversyl Plus, Coversyl Plus LD, Coversyl Plus HD, generics<br><br><\$20 | 2/0.625 mg, 4/1.25 mg or 8/2.5 mg once daily PO  | Combination of <a href="#">ACE</a> inhibitor and diuretic adverse effects possible, e.g., dry cough, electrolyte disturbances, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose). Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses). Monitor SCr and K <sup>+</sup> . Ineffective in patients with ClCr <30–40 mL/min. |
| <b>quinapril</b>  / <b>hydrochlorothiazide</b> <br>generics<br><br><\$20                                         | 10/12.5 mg, 20/12.5 mg or 20/25 mg once daily PO | Combination of <a href="#">ACE</a> inhibitor and diuretic adverse effects possible, e.g., dry cough, electrolyte disturbances,                             | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and                                                                                                                                                                                   | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b>                                                                                                                                                                                         |

| Drug/Cost <sup>[a]</sup>                                                                                                                                                                                                                              | Dosage                                                                | Adverse Effects                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                                                                                    | Comments                                                                                                                                                                                                                                                                                                 |
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|                                                                                                                                                                                                                                                       |                                                                       | hypotension, renal failure.                                                                                                                                | increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose).<br>Reduced efficacy of antihyperglycemic agents.                                                                                                                                                                                 | Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses).<br><br>Monitor SCr and K <sup>+</sup> .<br><br>Ineffective in patients with ClCr <30–40 mL/min.                                                                                                    |
| <b>ramipril</b>  / <b>hydrochlorothiazide</b> <br>Altace HCT, generics<br><br><\$20 | 2.5/12.5 mg, 5/12.5 mg, 10/12.5 mg, 5/25 mg or 10/25 mg once daily PO | Combination of <a href="#">ACE</a> inhibitor and diuretic adverse effects possible, e.g., dry cough, electrolyte disturbances, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose).<br>Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses).<br>Monitor SCr and K <sup>+</sup> .<br><br>Ineffective in patients with ClCr <30–40 mL/min. |

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| Drug Class: ARB/Diuretic Combinations                                                                                                             |                                                  |                                                                                                                                                      |                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                              |
| <b>azilsartan/chlorthalidone</b> <br>Edarbyclor<br><br>\$40–60 | 40/12.5 mg, 40/25 mg or 80/12.5 mg once daily PO | Combination of <a href="#">ARB</a> and diuretic adverse effects possible, e.g., electrolyte disturbances, hyperuricemia, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses). |

| Drug/Cost <sup>[a]</sup>                                                                                                                                                           | Dosage                                   | Adverse Effects                                                                                                                                      | Drug Interactions                                                                                                                                                                                                                                                                                                                                                    | Comments                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                    |                                          |                                                                                                                                                      | levels, adjust dose).<br>Reduced efficacy of antihyperglycemic agents.                                                                                                                                                                                                                                                                                               | Monitor SCr and K <sup>+</sup> .<br><br>Ineffective in patients with ClCr <30–40 mL/min.                                                                                                                                                                                                                 |
| <b>candesartan/hydrochlorothiazide</b> <br><a href="#">Atacand Plus</a> , generics<br><br><\$20 | 16/12.5 mg once daily PO                 | Combination of <a href="#">ARB</a> and diuretic adverse effects possible, e.g., electrolyte disturbances, hyperuricemia, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose).<br>Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses).<br>Monitor SCr and K <sup>+</sup> .<br><br>Ineffective in patients with ClCr <30–40 mL/min. |
| <b>irbesartan/hydrochlorothiazide</b> <br>Avalide, generics<br><br><\$20                        | 150/12.5 mg or 300/12.5 mg once daily PO | Combination of <a href="#">ARB</a> and diuretic adverse effects possible, e.g., electrolyte disturbances, hyperuricemia, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose).<br>Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses).<br>Monitor SCr and K <sup>+</sup> .<br><br>Ineffective in patients with ClCr <30–40 mL/min. |

| Drug/Cost <sup>t@a</sup>                                                                                                                                                            | Dosage                                           | Adverse Effects                                                                                                                                      | Drug Interactions                                                                                                                                                                                                                                                                                                                                                 | Comments                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>losartan/hydrochlorothiazide</b> <br>Hyzaar, Hyzaar DS, generics<br><br><\$20                   | 50/12.5 mg or 100/25 mg once daily PO            | Combination of <a href="#">ARB</a> and diuretic adverse effects possible, e.g., electrolyte disturbances, hyperuricemia, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose).<br>Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses).<br>Monitor SCr and K <sup>+</sup> .<br>Ineffective in patients with ClCr <30–40 mL/min. |
| <b>olmesartan/hydrochlorothiazide</b> <br>Olmotec Plus, generics<br><br><\$20                      | 20/12.5 mg, 40/12.5 mg or 40/25 mg once daily PO | Combination of <a href="#">ARB</a> and diuretic adverse effects possible, e.g., electrolyte disturbances, hyperuricemia, hypotension, renal failure. | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose).<br>Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses).<br>Monitor SCr and K <sup>+</sup> .<br>Ineffective in patients with ClCr <30–40 mL/min. |
| <b>telmisartan/hydrochlorothiazide</b> <br><a href="#">Micardis Plus</a> , generics<br><br><\$20 | 80/12.5 mg or 80/25 mg once daily PO             | Combination of <a href="#">ARB</a> and diuretic adverse effects possible, e.g., electrolyte disturbances, hyperuricemia,                             | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and                                                                                                                                                                                      | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b>                                                                                                                                                                                                  |

| Drug/Cost <sup>t@a</sup>                                                                                                                                                       | Dosage                                             | Adverse Effects                                                                                                                                           | Drug Interactions                                                                                                                                                                                                                                                                                                                                                 | Comments                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>valsartan/hydrochlorothiazide</b> <br><a href="#">Diovan-HCT</a> , generics<br><br><\$20 | 80/12.5 mg, 160/12.5 mg or 160/25 mg once daily PO | Combination of <a href="#">ARB</a> and diuretic adverse effects possible, e.g., electrolyte disturbances, hyperuricemia, hypotension, renal failure.      | Monitor serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or potassium-sparing diuretics. Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>Li <sup>+</sup> excretion may be altered (monitor Li <sup>+</sup> levels, adjust dose).<br>Reduced efficacy of antihyperglycemic agents. | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low diuretic doses).<br>Monitor SCr and K <sup>+</sup> .<br>Ineffective in patients with ClCr <30–40 mL/min. |
| Drug Class: Beta <sub>1</sub> -adrenergic Antagonist/Diuretic Combinations                                                                                                     |                                                    |                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                      |
| <b>atenolol/chlorthalidone</b> <br>generics<br><br><\$20–40                               | 50/25 mg or 100/25 mg once daily PO                | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams. Less common: hyperglycemia, depression, heart failure, heart block. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> .<br>Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone.<br><br>Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose).                                                                                                            | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated. Avoid in patients with asthma. <sup>[46]</sup><br><br>Avoid abrupt withdrawal (may precipitate rebound hypertension and                                                                  |



| Drug/Cost <sup>[a]</sup> | Dosage | Adverse Effects                                                                                                                                   | Drug Interactions                                                                                   | Comments                                                                                     |
|--------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                          |        | Hypotension, weakness, muscle cramps, impotence.                                                                                                  | <a href="#">NSAIDs</a> reduce hypotensive efficacy.                                                 | ischemia). Taper the dose before discontinuation.                                            |
|                          |        | Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.                                                                          | Diuretic-induced hypokalemia increases the risk of digoxin toxicity.                                | Avoid in patients with severe <a href="#">PAD</a> .                                          |
|                          |        | Rare: azotemia, blood dyscrasias, allergic reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity. | Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker. | Particularly effective in ISH, older and Black patients.                                     |
|                          |        |                                                                                                                                                   | Reduced efficacy of antihyperglycemic agents.                                                       | Monitor SCr and K <sup>+</sup> .                                                             |
|                          |        |                                                                                                                                                   |                                                                                                     | Consider alternatives in patients with or predisposed to arrhythmias.                        |
|                          |        |                                                                                                                                                   |                                                                                                     | Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low doses). |
|                          |        |                                                                                                                                                   |                                                                                                     |                                                                                              |
|                          |        |                                                                                                                                                   |                                                                                                     |                                                                                              |

|                                                                                                                        |                                    |                                                                                                                                                           |                                                                                                                                                             |                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>pindolol/hydrochlorothiazide</b>  | 10/25 mg or 10/50 mg once daily PO | Fatigue, bradycardia, decreased exercise capacity, headache, impotence, vivid dreams. Less common: hyperglycemia, depression, heart failure, heart block. | Bradycardia with digoxin or nondihydropyridine <a href="#">CCBs</a> . Cardiodepressant effects with nondihydropyridine <a href="#">CCBs</a> and amiodarone. | Beta-blockers should not be used as initial therapy in patients >60 y of age unless specifically indicated.         |
| <b>\$40–60</b>                                                                                                         |                                    |                                                                                                                                                           | Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose). <a href="#">NSAIDs</a> reduce hypotensive efficacy.                        | Avoid in patients with asthma. <sup>[46]</sup>                                                                      |
|                                                                                                                        |                                    |                                                                                                                                                           |                                                                                                                                                             | Avoid abrupt withdrawal (may precipitate rebound hypertension and ischemia). Taper the dose before discontinuation. |

| Drug/Cost <sup>[a]</sup> | Dosage | Adverse Effects                                                                                                                                   | Drug Interactions                                                    | Comments                                                                                            |
|--------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|                          |        | Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.                                                                          | Diuretic-induced hypokalemia increases the risk of digoxin toxicity. | Avoid in patients with severe <a href="#">PAD</a> .                                                 |
|                          |        | Rare: azotemia, blood dyscrasias, allergic reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity. | Reduced efficacy of antihyperglycemic agents.                        | Contraindicated in patients with second- or third-degree heart block in the absence of a pacemaker. |
|                          |        |                                                                                                                                                   |                                                                      | Particularly effective in ISH, older and Black patients.                                            |
|                          |        |                                                                                                                                                   |                                                                      | Monitor SCr and K <sup>+</sup> .                                                                    |
|                          |        |                                                                                                                                                   |                                                                      | Consider alternatives in patients with or predisposed to arrhythmias.                               |
|                          |        |                                                                                                                                                   |                                                                      | Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low doses).        |
|                          |        |                                                                                                                                                   |                                                                      |                                                                                                     |
|                          |        |                                                                                                                                                   |                                                                      |                                                                                                     |

Drug Class: Calcium Channel Blocker/ARB Combinations

|                               |                                                      |                                                                                                                                                                                                                                |                                                                                                                                                     |                                                                                                                                                                                                        |
|-------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>amlodipine/telmisartan</b> | 5/40 mg, 5/80 mg, 10/40 mg or 10/80 mg once daily PO | Ankle edema, flushing, headache, palpitations. Hyperkalemia. Can precipitate renal failure in susceptible patients (bilateral renovascular disease, those with volume depletion or with concurrent <a href="#">NSAID</a> use). | CYP3A4 substrate (many potential interactions). Strong inhibitors include azole antifungals, protease inhibitors, macrolides, quinidine.            | <b>Contraindicated in pregnancy—caution when prescribing to patients of childbearing potential.</b> <sup>[14]</sup>                                                                                    |
| <b>\$20–40</b>                |                                                      | Angioedema has been reported,                                                                                                                                                                                                  | Grapefruit juice may increase serum concentrations. Marked increase in serum K <sup>+</sup> in patients receiving K <sup>+</sup> supplements and/or | Use lower initial doses in patients who are volume depleted or on diuretics (increased risk of hypotension in hypovolemia). Hyperkalemia usually occurs only in those on K <sup>+</sup> supplements or |

| Drug/Cost <sup>[a]</sup>                                                            | Dosage                                                                    | Adverse Effects                                                                                                                                         | Drug Interactions                                                                                                                                                                                                                                                        | Comments                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                     |                                                                           | but a causal association has not been established.                                                                                                      | potassium-sparing diuretics.<br>Reduced hypotensive effect with <a href="#">NSAIDs</a> and increased risk of renal dysfunction.<br><br>May elevate Li <sup>+</sup> levels (monitor Li <sup>+</sup> levels, adjust dose).                                                 | drugs that cause K <sup>+</sup> retention, those with renal impairment or diabetics with high serum K <sup>+</sup> levels.<br>Assess SCr and K <sup>+</sup> after a few days, then regularly. |
| Drug Class: Calcium Channel Blocker/HMG-CoA Reductase Inhibitor Combinations        |                                                                           |                                                                                                                                                         |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                               |
| <b>amlodipine/atorvastatin</b><br><a href="#">Caduet</a> , generics<br><br><\$20–40 | Amlodipine 5 or 10 mg plus atorvastatin 10, 20, 40 or 80 mg once daily PO | Ankle edema, flushing, headache, palpitations.<br>Adverse effects of atorvastatin include constipation, flatulence, dyspepsia, abdominal pain, myalgia. | CYP3A4 substrate (many potential interactions).<br>Strong inhibitors include azole antifungals, protease inhibitors, macrolides, quinidine.<br><br>Grapefruit juice may increase serum concentrations.<br><br>Amlodipine and atorvastatin are both substrates of CYP3A4. | For patients with hypertension and an indication for an HMG-CoA inhibitor.                                                                                                                    |

|                                                                                                                                                                                                                                              |                               |                                                                                                                                                                       |                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                   |
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| Drug Class: Diuretic Combinations                                                                                                                                                                                                            |                               |                                                                                                                                                                       |                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                   |
| <b>hydrochlorothiazide /amiloride  (50/5)</b><br>generics<br><br><\$20 | One-half tablet once daily PO | Hypotension, weakness, muscle cramps, impotence.<br>Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.<br><br>Rare: azotemia, blood dyscrasias, | Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose).<br><a href="#">NSAIDs</a> reduce hypotensive efficacy.<br><br>Diuretic-induced hypokalemia increases the risk of digoxin toxicity.<br><br>Reduced efficacy of antihyperglycemic | Particularly effective in ISH, older and Black patients.<br>Monitor SCr and K <sup>+</sup> .<br><br>Consider alternatives in patients with or predisposed to arrhythmias.<br><br>Can exacerbate gout and diabetes |

| Drug/Cost <sup>[a]</sup>                                                                                                                                    | Dosage                                                               | Adverse Effects                                                                                                                                                                                                                                                                       | Drug Interactions                                                                                                                                                                                                                                                                                                                                  | Comments                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                             |                                                                      | allergic reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity.                                                                                                                                                                       | agents.<br><br>May exacerbate <a href="#">ACE</a> inhibitor–induced hyperkalemia.                                                                                                                                                                                                                                                                  | (biochemical abnormalities are less frequent at low doses).<br><br>Lower incidence of hypokalemia than with hydrochlorothiazide alone.                                                                                                                                                                                                                   |
| <b>hydrochlorothiazide /triamterene (25/50)</b><br>generics<br><br><\$20 | Initial: One-half tablet<br><br>Usual: 1 tablet<br><br>Once daily PO | Hypotension, weakness, muscle cramps, impotence.<br>Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.<br><br>Rare: azotemia, blood dyscrasias, allergic reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity. | Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose).<br><a href="#">NSAIDs</a> reduce hypotensive efficacy.<br><br>Diuretic-induced hypokalemia increases the risk of digoxin toxicity.<br><br>Reduced efficacy of antihyperglycemic agents.<br><br>May exacerbate <a href="#">ACE</a> inhibitor–induced hyperkalemia. | Particularly effective in ISH, older and Black patients.<br>Monitor SCr and K <sup>+</sup> .<br><br>Consider alternatives in patients with or predisposed to arrhythmias.<br><br>Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low doses).<br><br>Lower incidence of hypokalemia than with hydrochlorothiazide alone. |

|                                                                                                                                                                                                                                                        |                                                                      |                                                                                                                                                                       |                                                                                                                                                                                                                     |                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>hydrochlorothiazide /spironolactone  (25/25)</b><br>generics<br><br><\$20 | Initial: One-half tablet<br><br>Usual: 1 tablet<br><br>Once daily PO | Hypotension, weakness, muscle cramps, impotence.<br>Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia, hyperlipidemia.<br><br>Rare: azotemia, blood dyscrasias, | Li <sup>+</sup> excretion reduced (monitor Li <sup>+</sup> levels, adjust dose).<br><a href="#">NSAIDs</a> reduce hypotensive efficacy.<br><br>Diuretic-induced hypokalemia increases the risk of digoxin toxicity. | Particularly effective in ISH, older and Black patients.<br>Monitor SCr and K <sup>+</sup> .<br><br>Consider alternatives in patients with or predisposed to arrhythmias. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Drug/Cost <sup>[a]</sup> | Dosage | Adverse Effects                                                                                                 | Drug Interactions                                                                                                       | Comments                                                                                                                                                                |
|--------------------------|--------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |        | allergic reactions (potential cross sensitivity with other sulfonamide derivatives), fatigue, photosensitivity. | Reduced efficacy of antihyperglycemic agents.<br><br>May exacerbate <a href="#">ACE</a> inhibitor-induced hyperkalemia. | Can exacerbate gout and diabetes (biochemical abnormalities are less frequent at low doses).<br><br>Lower incidence of hypokalemia than with hydrochlorothiazide alone. |
|                          |        | Gynecomastia in men and breast tenderness in women.                                                             |                                                                                                                         |                                                                                                                                                                         |

<sup>[a]</sup> Cost of 30-day supply of usual dose of drug; includes drug cost only.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).  
**Abbreviations:** ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; AV = atrioventricular; CCB = calcium channel blocker; CNS = central nervous system; CV = cardiovascular; DBP = diastolic blood pressure; GI = gastrointestinal; ISA = intrinsic sympathomimetic activity; ISH = isolated systolic hypertension; NSAID = nonsteroidal anti-inflammatory drug; PAD = peripheral arterial disease; SBP = systolic blood pressure; SCr = serum creatinine

## Suggested Readings

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[Hypertension Canada recommendations on the management of hypertension are updated regularly. A summary of the important and new recommendations can be found in the \*Healthcare Professionals\* area and is also broadly published in multidisciplinary journals.](#)

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